Customer Journey Map

Date	06-07-2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Customer Journey Map

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

Phase of Journey	**Discovery & Data Input**	**Prediction & Insight Review**	**Tracking & Continued Optimization**
Actions *What does the customer do?*	• Lands on the OptiCrop homepage and reads the value proposition • Reviews accepted input ranges (N, P, K, temperature, humidity, pH, rainfall) • Enters soil and climate readings into the Crop Recommendation form	• Submits the form (POST /predict) to request a recommendation • Views the predicted crop, confidence score, and probability breakdown • Checks season, water need, and fertilizer guidance; downloads PDF report	• Opens Prediction History to review past recommendations • Exports history as CSV or downloads past PDF reports • Visits the Dashboard to check model accuracy and crop distribution
Touchpoint *What part of the service do they interact with?*	• OptiCrop landing page (index.html) and Quick Input info card • Crop Recommendation web form (7 numeric input fields) • Flask route: GET / and POST /predict form submission	• Result page (result.html) rendered by the /predict route • ML pipeline: model.pkl, scaler.pkl, label_encoder.pkl (model.py) • PDF report generator (FPDF) via the /download route	• History page (history.html) and /history?format=csv export • Dashboard page (dashboard.html) with accuracy & crop charts • SQLite database (database.db) via database.py / SQLAlchemy
Customer Thought *What is the customer thinking?*	• "Will this match my actual soil and climate conditions?" • "I hope the input ranges are clear so I enter correct values" • "This form looks quick, but will the result be accurate?"	• "Is this recommended crop really the best choice for me?" • "How confident is the model in this prediction?" • "I want a report I can save or show to others"	• "Are my earlier recommendations still useful this season?" • "How reliable is this model overall compared to others?" • "I want to keep my history clean and relevant"
Customer Feeling *What is the customer feeling?*	Curious but slightly unsure about entering technical soil readings correctly.	Reassured and confident once a clear crop, confidence score, and guidance appear.	Engaged and in control, satisfied with visibility into past predictions and model performance.

Process Ownership *Who is in the lead on this?*	Frontend/UI team (Flask templates, static assets) and product owner responsible for form design and input guidance.	ML/Data Science team (model.py, train_model.py) together with the Backend team (app.py) for prediction accuracy and report generation.	Backend/Database team (database.py, SQLAlchemy models) and the analytics owner maintaining the dashboard and history views.
Opportunities *How can we improve this stage?*	• Add inline tooltips/units for each soil and climate field • Provide sample or auto-filled values to reduce guesswork • Add inline validation messages before submission	• Show which features most influenced the prediction • Add a confidence warning when probability is low • Improve PDF report styling and add charts	• Add trend charts comparing predictions across seasons • Allow bulk export/delete and filtering of history records • Send reminders to re-run predictions each growing season